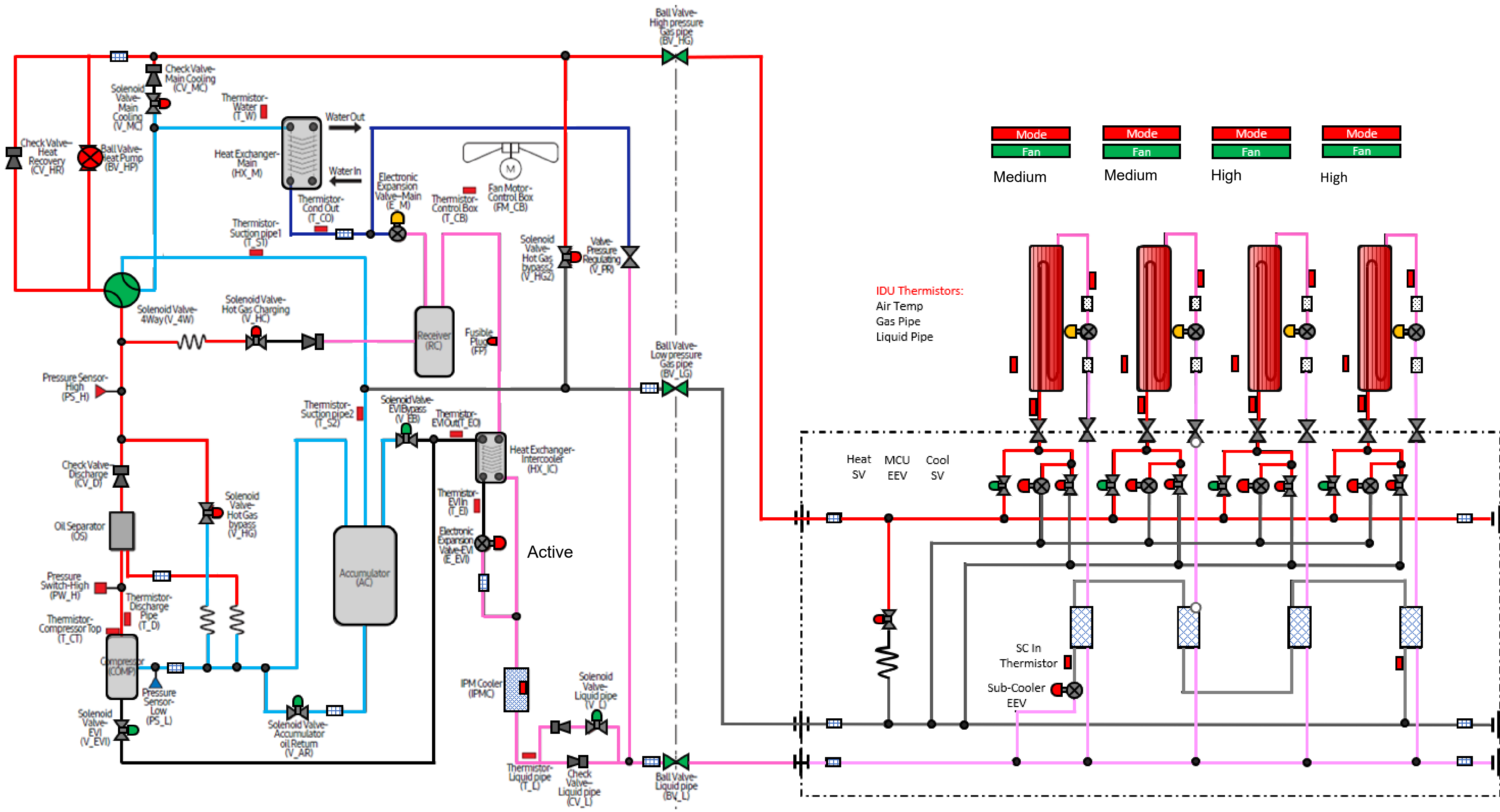
SAMSUNG

DVM S 3 Phase Water Heat Recovery
Heating Only Mode



Comments:

Outdoor Unit			
Component	**Target**	**Reading**	
Water Out Temp.			
Compressor Hz.			
High pressure	407 psig		Ok
H.P Sat. Temp			
Discharge Temp.	150^0-212^0		Ok
IPM Temp,	Below 194^0		
Low pressure			Low
L.P. Sat. Temp	2-10^0 Below T_W		
Suction 1 Temp.			Ok
Suction 2 Temp			
S.H.	1^0 - 12.6^0	***0***	Ok
Min EEV Position	Modulating		Ok
Cond. Out Temp.	2-10^0 Below T_W		Ok
Liquid Line Temp.			Ok
Sub-Cooling @ T_L		***0***	Ok

Indoor Units			
Component	**Target**	**Reading**	
IDU 1 EEV	500 – 1500 Steps		Ok
IDU 1 SC	9^0	***0***	Ok
IDU 1 T PI			Ok
IDU 1 T PO			Ok
IDU 2 EEV	500 – 1500 Steps		Ok
IDU 2 SC	9^0	***0***	Ok
IDU 2 T PI			Ok
IDU 2 T PO			Ok
IDU 3 EEV	500 – 1500 Steps		Ok
IDU 3 SC	9^0	***0***	Ok
IDU 3 T PI			Ok
IDU 3 T PO			Ok
IDU 4 EEV	500 – 1500 Steps		Ok
IDU 4 SC	9^0	***0***	Ok
IDU 4 T PI			Ok
IDU 4 T PO			Ok

Directions:
Enter the data in the spaces provided. The reading will populate in the refrigerant diagram. Space temperature is entered at the IDU. Use the drop down boxes to identify the state of the component. Use the operating data and the target operation data on the next page to troubleshoot the system.

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DVM S 3 Phase Water Heat Recovery
Heating Only Target Operation

Water Source Outdoor Unit

Component	Abbrevaition		Heating Only Mode	Comments
Compressor	Comp	Operation	Target High pressure control (default: 407 psig)	Primary function is to maintain head pressure (range 356-455psig)
		Range	20~160Hz	
Main EEV	E_M	Operation	HR - Target SH = Water temperatrue - 5.4°F	HP - Target SH = (Low pressure-10)*7+5
		Range	125 - 2000 Steps	Suction Superheat 1.8°F - 12.6°F
EVI EEV	E_EVI	Operation	On (Open)	By-Pass Mode 18°F Superheat Between pipe in and pipe out
		Range	0-480step	EVI Vapor Injection Mode 3.6°F Superheat Between pipe in and pipe out
4 Way Valve	V_4W	Operation	On (Energized)	
EVI By-Pass Solenoid Valve	V_EB	Operation	On (Open): *T_W > 50.4 or 50Hz↓	By-pass mode, Discharge Temperature control
			Off (Close): T_W < 45°F ↓ and 65Hz↑	Low Ambieant heating Operation Vapor Injection circuit
EVI Solenoid Valve	V_EVI	Operation	On (Open)	
Hot Gas By-Pass	V_HG	Operation	Off (Closed)	
Accumulator Oil Return Solenoid Valve	V_AR	Operation	On (Open)	**Off** during start up if DSH is < 18°F. **Off** for 2 min. if DSH is < 18°F **On** when DCSH > 27°F
Hot Gas By-Pass 2 (HR Only)	V_HG2	Operation	Off Closed	
Liquid Line Solenoid Valve	V_L	Operation	On (Open)	
Main Cooling Solenoid	V_MC	Operation	Off (Closed)	
Hot Gas Charging Solenoid Valve	V_HC	Operation	Off (Closed)	
Condenser Out Thermistor	T_CO	Value	Should be 2 degrees lower than the water out temperature	Low Side saturation temperature is ***"typically"*** 2-10 degrees lower than water temperature

Mode Change Unit (MCU)

Component	Abbrevaition		Heating Mode	Comments
Port Heating Solenoid	Heat	Operation	On (Open) Call for Heating Off (Closed) Call for Cooling	Energized for units calling for heating
Port Cooling Solenoid	Cool	Operation	On (Open) Call for Cooling Off (Closed) Call for Heat	Energized for units calling for cooling
Port EEV	EEV	Operation	Closed 0 Steps	Open 480 step when a zone calls for cooling or during a mode change.
		Range	0 - 480 Steps	Closed 0 Steps when a zone is in heating
Sub_Cooling EEV	SC_EEV	Operation	Active inheating Main	Only active in mixed mode of operation
		Range	Not active	
Sub-Cooler In Temperature		Operation	Should be colder then Pipe Out	
Sub-Cooler Out Temperature		Operation	Should be hotter then pipe in.	

Indoor Unit IDU

Component	Abbrevaition		Heating Mode	Comments
Room Temperature	Room Temp.	Operation	67-80 Degrees	
Liquid Pipe Thermistor	EVA In 1	Operation	Liquid Pipe Temperature for sub_cooling Control	Used to calculate Evaporator Sub-Cooling
Gas Pipe Theromistor	EVA Out 1	Operation	Hot Gas Pipe Temperature should be close to discharge temp.	Used to evaluate the temperature of the hot gas coming into the coil
EEV	EEV 1	Operation	Modulates to maintain 9 degrees of sub-cooling	Sub-Cooling = High Side Saturation Temperature - Pipe in Temp.
		Range	500- 1500 Steps (80 - 120 Steps when zone is satisfied)	Typical range between 500 and 1500 Steps
IDU Fan	Fan Speed	Range	Low , Medium, High, Auto	
		Operation	Low Speed: $\lvert Ts - Ti \rvert \le 1.8$°F	TS - Set Temperature
			Medium Speed: 1.8°F < $\lvert Ts - Ti \rvert \le 5.4$°F	Ti Indoor Temperature
			High Speed 5.4°F < $\lvert Ts - Ti \rvert$	

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